

A Comparative Study on Cardiac Ganglia in Midday Gerbil, Egyptian Spiny Mouse, Chinchilla Laniger and Pigeon

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With 4 figures

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Summary

Using the thiocholine method and histological techniques, the topography and morphology of cardiac ganglia in midday gerbil, Egyptian spiny mouse, chinchilla laniger and pigeon were studied. The results demonstrated that cardiac ganglia in all investigated species are embedded in epicardial fat. They formed plexo-ganglionic structures. Each of them composed of many ganglia (from seven up to 36) different in size and shape, and interconnected by fascicles of nerve fibres. Comparative analysis showed that the density of neural network and cell aggregations was different in individual species. The richest plexo-ganglionic structure was in pigeon. It was organized in three plexo-ganglia with an average of 30 ganglia. The largest one was located along the anterior interventricular sulcus. The cardiac ganglia of investigated mammals were localized mainly on the epicardium of atria; in Egyptian spiny mouse and chinchilla laniger on the ventral surface of right atrium, but in midday gerbil on the dorsal surface of left atrium. Moreover, in midday gerbil and Egyptian spiny mouse the little plexo-ganglionic structure on the ventricle were noticed. Additionally, in midday gerbil the single nerve cells might be observed between cardiac muscle of atria. It can be said that, the strongly developed cardiac plexus in pigeon is probably connected with his behaviour and functional properties of the heart. The arrangement of neurones in cardiac ganglia of all examined mammals was uniform over the whole surface of the sections, while in the pigeon, neurones were located mainly in the peripheral part of the ganglion.

Introduction

The cardiac ganglia have been examined and described in many species of amphibians, reptiles, birds and mammals. The research were concerned with their morphology, topography, cytoarchitectonics, ultrastructure and also physiology of cardiac neurones. The results of these researches revealed that the cardiac ganglia were found in a number of specific cardiac regions (Michell, 1956; Bojsen-Moller and Traum-Jensen, 1971; Randall et al., 1986; Baluk and Gabella, 1987; Hancock et al., 1987; Armour and Hopkins, 1990; Furukawa et al., 1990; Pauza et al., 1997, 1999, 2000, 2002; Kuder and Tekieli, 1999, 2000; Pauziene et al., 2000).

They formed a system of ganglionated plexuses of diverse size, shape and density. The highest density of epicardiac ganglia was found on the surfaces of the left and right atria. Moreover, the ganglia were located on both surfaces of cardiac ventricle.

The classical view regarding cardiac ganglia is that they contain parasympathetic efferent post-ganglionic neurones (Bojsen-Moller and Traum-Jensen, 1971; Ellison and Hibbs, 1976; Baluk and Gabella, 1987; Hancock et al., 1987; Michell, 1956). On the other hand, the presence of a few adrenergic neurones in the cardiac ganglia was identified by Armour and Hopkins, (1990) and Yuan et al. (1994). Some authors demonstrated morphological differentiation of neurones in the cardiac plexuses and they thought, that the function of the neurones may also be different (Edwards et al., 1995; Pauza et al., 1997).

Nevertheless, there is no thorough study of the cardiac ganglia in chinchilla laniger, Egyptian spiny mouse, midday gerbil and pigeon – except the congress abstracts (Kuder et al., 2000a,b; 2001). The present examinations concerning the topography, morphology and cytoarchitectonics in Egyptian spiny mouse, chinchilla laniger, midday gerbil and pigeon has been undertaken to complete data and carry out the comparative analysis.

Materials and Methods

Eight adult Egyptian spiny mice (*Acomys cahirynus*), five chinchilla lanigers (*Chinchilla laniger*), seven midday gerbils (*Meriones meridianus*) and eight pigeons (*Columba livia*) of either sex have been used in the study. The animals were deeply anaesthetized with ether and intraperitoneal injection of Nembutal (20–40mg/kg). Then the hearts were taken out.

Histochemical study

Part of hearts were fixed in 10% formalin and then treated by a whole-mount technique using the histochemical method of Koelle and Friedenwald (1949) modified by Gienc (1976, 1977) for use in macromorphological specimens.

Histological study

Other hearts were fixed in 4% neutral formalin, then embedded in paraffin and then cut on a microtome into 5–8 µm sections. The hearts were sectioned from the base of the heart. The sections were stained with haematoxylin and eosin and Nissl's method.

Results

Midday gerbil

Histochemical investigations revealed the presence of delicate plexo-ganglionic structures in the fatty tissue of gerbil's

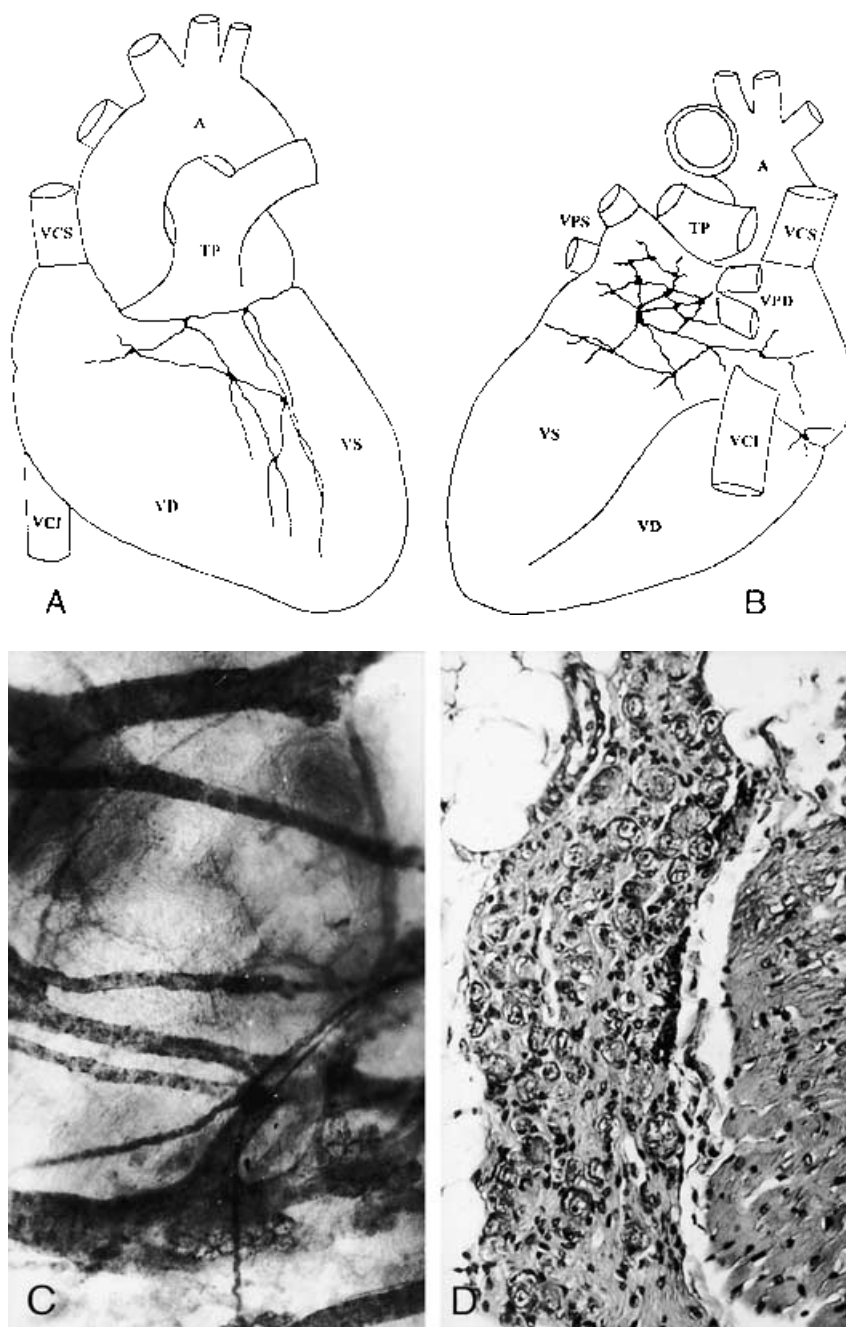


Fig. 1. Cardiac ganglia in midday gerbil: (A) scheme of distribution on the ventral surface; (B) scheme of distribution on the dorsal surface; (C) the part of plexo-ganglion from ventral surface of right atrium, thiocholine method, magn. approx. 120 $\times$ and (D) longitudinal section through the big ganglia from the left atrium, Nissl's method, magnification approx. 220 $\times$. Abbreviations: A, aorta; AD, atrium dextrum; AS, atrium sinistrum; TP, truncus pulmonalis; VCC, vena cava caudalis; VCCD, vena cava cranialis dextra; VCCS, vena cava cranialis sinistra; VCI, vena cava inferior; VCS, vena cava superior; VCSD, vena cava superior dextra; VD, ventriculus dexter; VPD, vena pulmonaris dextra; VPS, vena pulmonalis sinistra; VS, ventriculus sinister.

epicardium. It composed of about 20 ganglia interconnected by fascicles of nerve fibres. The measurement of ganglia ranged from 55 μm up to 220 μm in length and from 35 μm up to 105 μm in width. The plexo-ganglion was located on dorsal surface of heart – mainly on the left atrium. Moreover a few small ganglia in the vicinity of dorsal coronary sulcus and superior vena cava were observed (Fig. 1B). Another plexus, but less organized was located on the right cardiac ventricle (ventral surface). It contained one to two bigger ganglia (550 μm in length and 120 μm in width) and two to five smaller ones (Fig. 1A,C). The arrangement of nerve cells in all ganglia was uniform over the whole surface of the sections. Their diameter ranged from 17 to 36 μm (most often 26 μm). All the components of described plexuses were joined by nerve fibre

fascicles. Single cells could be observed along these fibres. The junction between the vagal nerve and heart were noticed.

The histological investigations confirm the presence of cell agglomerations connected by the fascicle of nerve fibres in the fatty tissue of left atrium and right ventricle (Fig. 1D). Moreover, the single nerve cells might be observed between cardiac muscle of atria.

Egyptian spiny mouse

In all investigated animals, the presence of two plexuses in the fat tissue of epicardium were observed (Fig. 2A,B,C). One of these was located on the dorsal surface of the left cardiac ventricle in the vicinity of the dorsal interventricular sulcus.

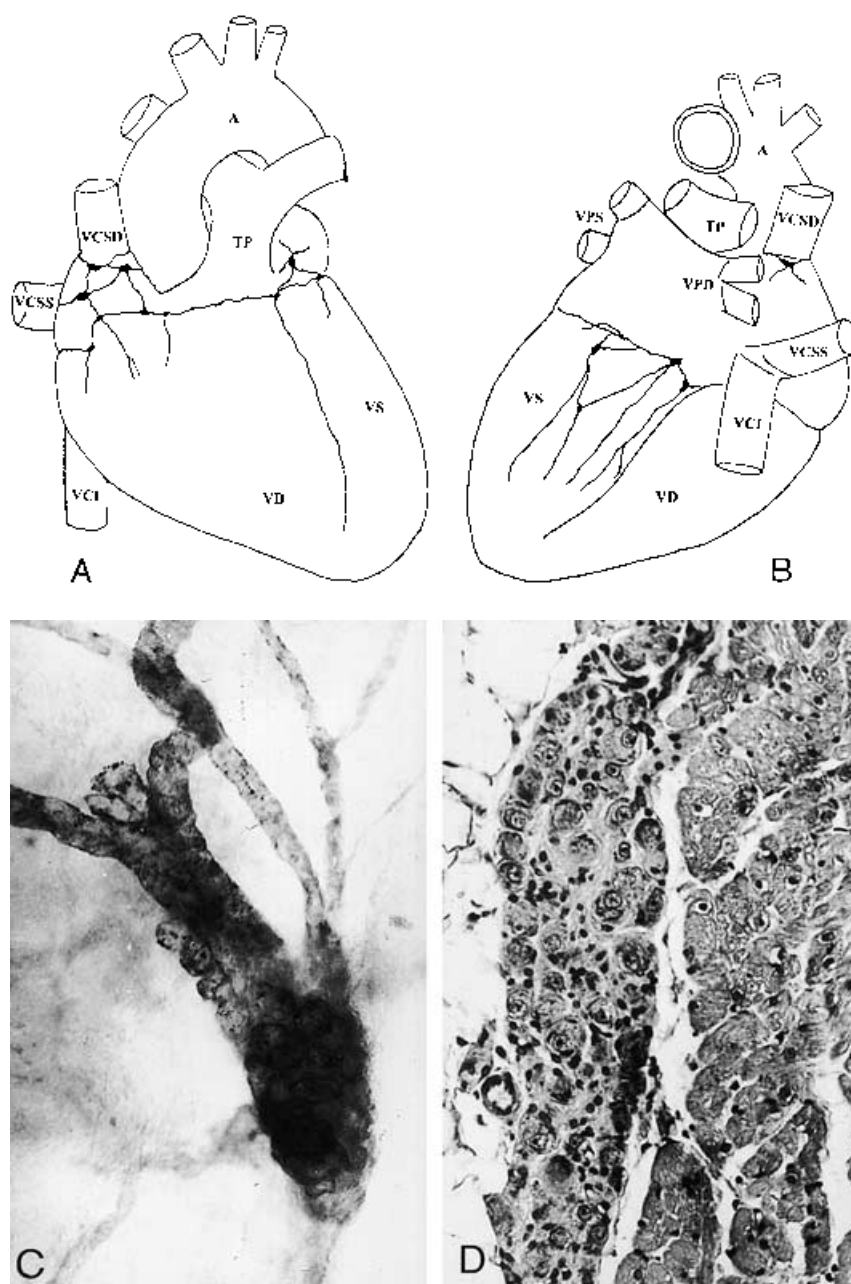


Fig. 2. Cardiac ganglia in Egyptian spiny mouse: (A) scheme of distribution on the ventral surface; (B) scheme of distribution on the dorsal surface; (C) the biggest ganglion from right atrium, thiocholine method, magnification approx. 150 $\times$ and (D) longitudinal section through the ganglion from the right atrium, Nissl's method, magnification approx. 220 $\times$. Abbreviations as in Fig. 1.

It usually consisted of four to five thick nerve fibre fascicles and two to three small ganglia near the coronary sulcus. These ganglia were 85 μ m in length and 50 μ m in width.

The other plexus was located in the fat tissue of the epicardium on the right atrium (ventral surface) (Fig. 2C,D). It consisted of three to four larger ganglia (550 μ m $\times$ 350 μ m and 10–15 cells on cross sections) and number of smaller ganglia consisting of a few cells. They were in compact arrangement and their diameter ranged from 18 to 38 μ m. All the ganglia were connected by nerve fibres and fascicles forming a plexo-ganglionic structure. Single nerve cells could be observed along these fibres. In two cases, a small ganglion at the left atrium was observed as well and a delicate complex of nerve fibres and fascicles on the right ventricle was also noticed. We observed delicate nerve fibres from vagal nerve, which were running to atrial ganglia.

Chinchilla laniger

In all animals, plexo-ganglionic structure with numerous ganglia and nerve fibres were detected. The biggest plexo-ganglion was usually located on the ventral surface of the heart (Fig. 3A,C). It consisted of two or three bigger ganglia on the right atrium, near the entry of superior vena cava, and numerous small ganglia in the vicinity of the coronary sulcus. The measurement of big ganglia was about 700 μ m in length and 450 μ m in width. The small ganglia were of various dimensions – from 50 μ m up to 295 μ m in length. Another plexus with one or two ganglia was observed on the left atrium. They were located in the depression between the right cardiac auricles and the aorta. Moreover, a few small ganglia on the dorsal surface of ventricles, near the coronary sulcus, were found (Fig. 3B,D). The results of histological

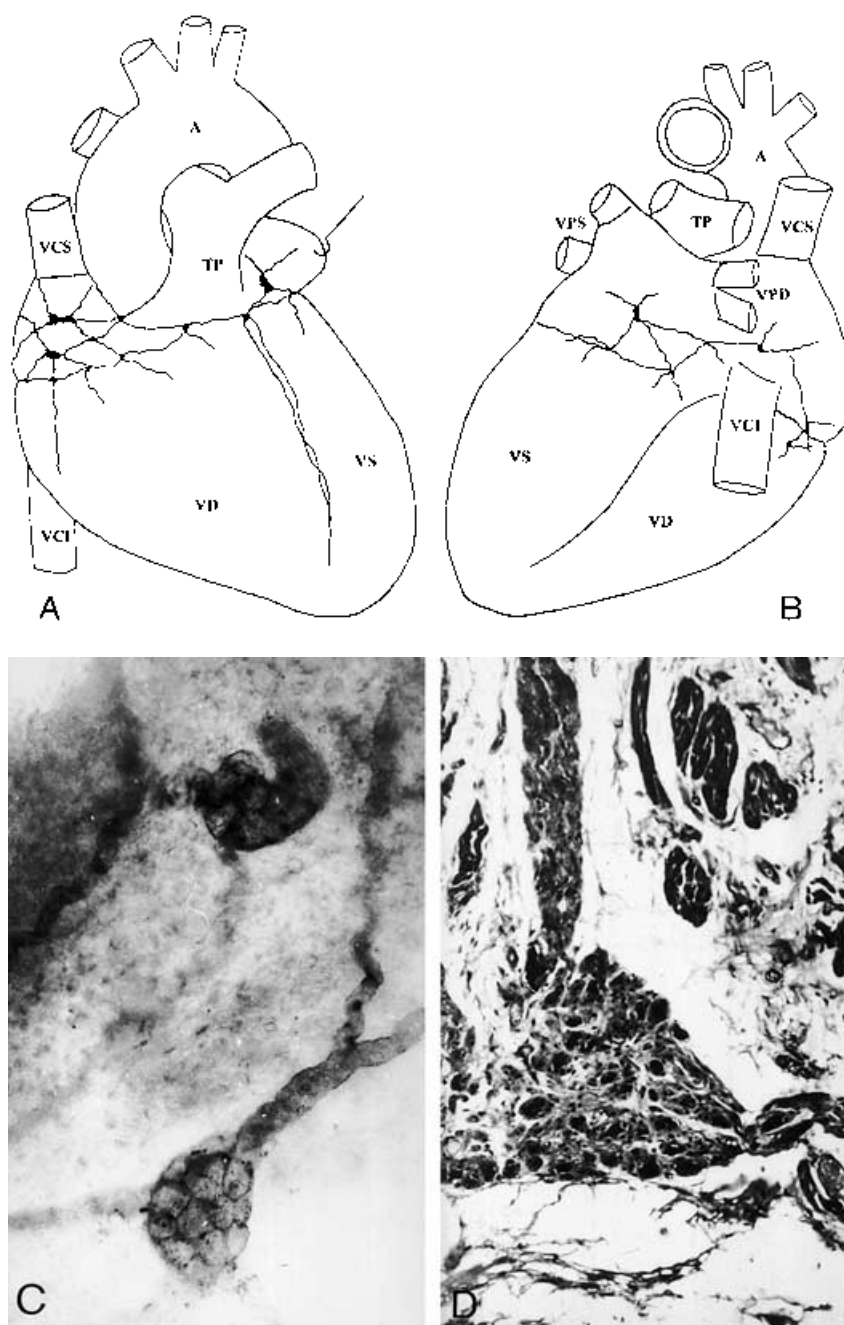


Fig. 3. Cardiac ganglia in chin-chilla laniger: (A) scheme of distribution on the ventral surface; (B) scheme of distribution on the dorsal surface; (C) two small ganglia from the ventral surface of right ventricle, thiocholine method, magnification approx. 180 $\times$ and (D) cross section through the ganglion from the dorsal surface of left atrium, H & E, magnification approx. 150 $\times$. Abbreviations as in Fig. 1.

investigations confirmed the presence of ganglionated plexus on the ventral and dorsal walls of atria. The arrangement of the neurones in all ganglia was uniform over whole surface of the sections. The diameter of ganglionic cells ranged from 15 to 35 μm .

Pigeon

Three plexo-ganglionic structures were observed in the fat tissue of pigeon's epicardium. One of them was located along the ventral coronary sulcus and the superior part of the ventral interventricular sulcus (Fig. 4A). It composed of about fifteen small ganglia (average 78 μm in length) and several nerve fascicles. The second plexo-ganglion was located on the dorsal surface of right and left atria (Fig. 4B). It consisted of three or

five big ganglia (80 μm –356 μm length), each composed of a few dozen ganglionic cells, and numerous small ones. The arrangement of neurones in big ganglia of pigeon was characteristic. They were grouped mainly in the peripheral part of the ganglia, while the nerve fibres were in the central part of the ganglia (Fig. 4D). The third plexo-ganglion was located on the dorsal surface of the cardiac ventricle (Fig. 4B,C). It consisted of numerous (10–16) small ganglia of various dimensions (54 μm to 132 μm in length and 38 μm to 86 μm in width) connected with each other by nerve fibres. The ganglia were located near the coronary sulcus and superior part of left ventricle. The histological sections showed, that in this species the ganglionic cells were located mainly in the external parts of ganglia, whereas the central parts were occupied by nerve fibres (Fig. 4D).

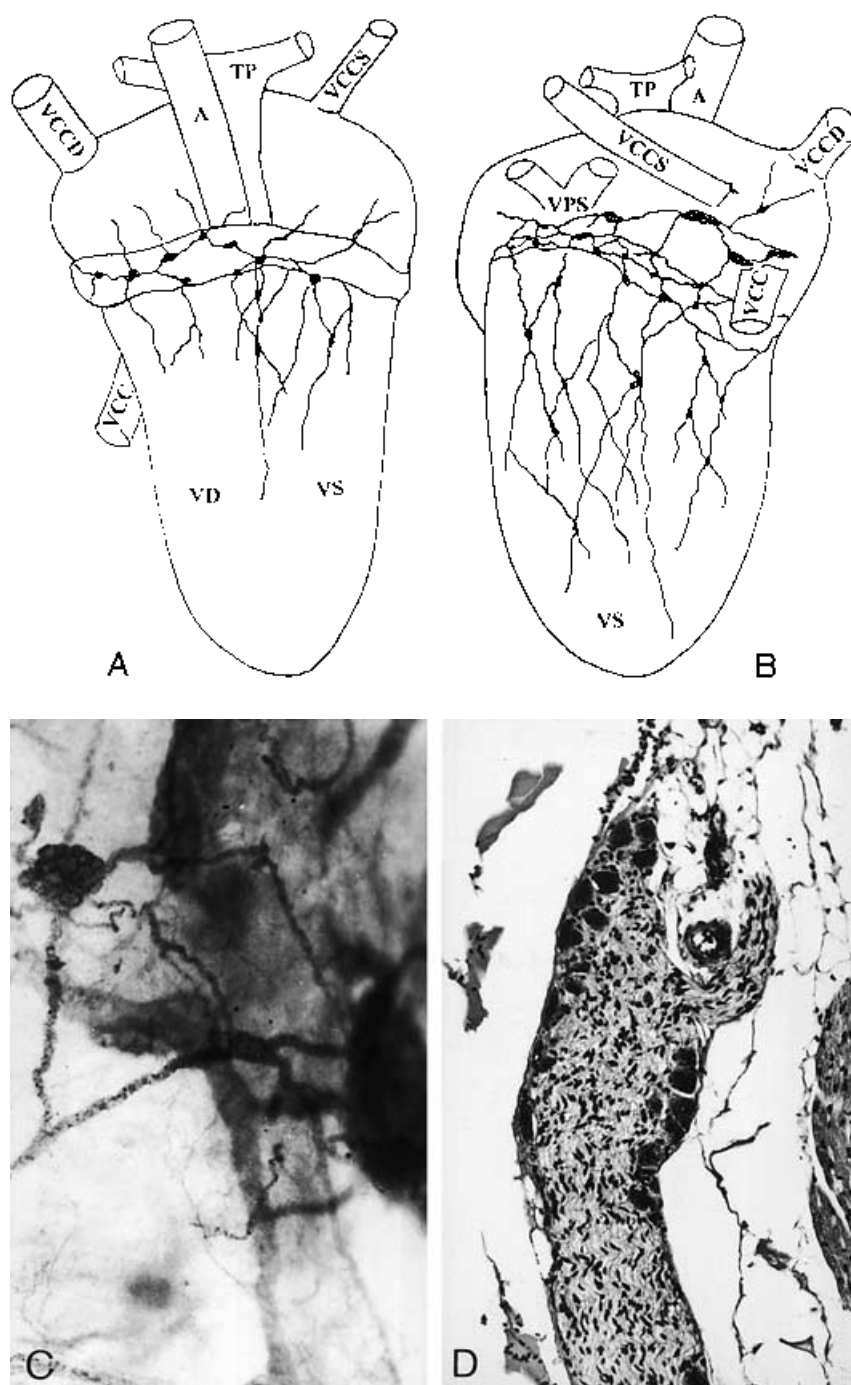


Fig. 4. Cardiac ganglia in pigeon: (A) scheme of distribution on the ventral surface; (B) scheme of distribution on the dorsal surface; (C) part of plexo-ganglion from the coronary sulcus of dorsal surface, thiocholine method, magnification approx. 100 $\times$ and (D) longitudinal section through the ganglion from ventral surface of right atrium, H & E, magnification approx. 200 $\times$. Abbreviations as in Fig. 1.

Discussion

The results of the histochemical and the histological studies demonstrated that cardiac ganglia in all investigated species are embedded in epicardial fat. Comparative analysis showed, that the density of neural network and cell aggregations in pigeon is bigger than in mammals. It was organized in three plexo-ganglionic structures with about 30–40 ganglia which were diverse in size and shape. The largest plexo-ganglion was located along the anterior interventricular sulcus. A similar organization of cardiac ganglia was reported by Kuder and Tekieli (1999) in Japanese quail, whereas Langenfeld (1992) described five cardiac plexuses in birds: (1) *plexus cardiacus*

cranialis dexter, (2) *plexus cardiacus cranialis sinister*, (3) *plexus cardiacus ventralis dexter*, (4) *plexus cardiacus ventralis sinister*, and (5) *plexus atrium ventralis*. On the basis of the above-mentioned data it can be said that the strongly developed cardiac plexuses in birds is probably connected with their behaviour and functional properties of the heart.

Some authors thought that cardiac ganglia appeared only in the epicardium of atria. However, the present study and the data in literature revealed the presence of small autonomic ganglia or autonomic plexuses on the ventricle wall and in the fat tissue of the coronary sulcus (Anderson and Smith, 1971; Ellison and Hibbs, 1976; Randal et al., 1986; Armour and

Hopkins, 1990; Yuan et al., 1993, 1994; Pauza et al., 1997, 1999, 2000; Kuder and Tekieli, 1999, 2000). Pauza et al. (2000) distinguished in human hearts, a system of seven-ganglionated subplexuses; the right atrium was innervated by two subplexuses, the left atrium by three, the right ventricle by one, and the left ventricle by three subplexuses. The highest density of these plexuses in human hearts was identified on the dorsal and dorsolateral surfaces of left atrium, where up to 50% of all cardiac ganglia were located (Pauza et al., 2000).

In midday gerbil the single nerve cells might be observed between cardiac muscle of atria. The same localization was observed by Michell (1956).

Pauza et al. (1997) revealed that, in the heart of rats and guinea pigs, there were two types of neurones: the unipolar and multipolar ones. On the other hand Edwards et al. (1995) distinguished three types of neurones in guinea pig cardiac ganglia of different electric potential. The authors suggested that these two or three types of cardiac neurones may play different roles.

The cardiac ganglia found in many species of vertebrates varied in size and number (Michell, 1956; Yuan et al., 1994; Kuder and Tekieli, 2000; Pauza et al., 2002). The human heart consist of an average of 836 ganglia (Pauza et al., 2000), the canine – 260 (Yuan et al., 1994). In described smaller species, there were about 30 cardiac ganglia. It suggested that the number of cardiac ganglia may be correlated with the body sizes and activity of the heart.

An interesting problem is the arrangement of neurones in cardiac ganglia. In three examined mammals it was uniform over the whole surface of the sections, while in the pigeon, mainly in the peripheral part of the ganglion. However such arrangement of cardiac neurones is not characteristic in another species of birds (Kuder and Tekieli, 2000).

Based on the data in literature and this study, we can say that in all examined animals and humans the diameters of neurone perikarya were similar and ranged from 15 μm up to 38 μm .

Cardiac ganglia were considered as parasympathetic, which does not rule out the presence of adrenergic cells. The parasympathetic nature of these ganglia was confirmed, among other things (Randal et al., 1986; Gatti et al., 1995), by histochemical investigations of Jacobovitz (1967), Abraham (1969), Taylor and Smith (1971), Edwards et al. (1995), Becker et al. (1997). Moreover, electron-microscopic research of Ellison and Hibbs (1976) revealed the presence of mainly light, electron-transparent vesicles in synapses. Also in the synaptic profiles in quail cardiac neurones in general, small, agranular vesicles were observed, but among them, the large ones with a dense core were noticed too (Kuder and Tekieli, 1999). The authors suggested that this kind of synaptic vesicles might represent neuropeptides.

Edwards et al. (1995) observed in guinea pig's heart bundles of nerve fibres leaving the vagal nerve and running to atrial ganglia. In our investigations, we found similar junctions.

It can be said that, despite the general similarity in topography and morphology, cardiac ganglia have some species-specific features. In the pigeon, such a feature is the presence of three ganglionated plexuses, while in all investigated mammals there are only two plexuses. Moreover, in this species the ganglionic cells were located mainly in the external part of ganglia, whereas the central part of ganglia was occupied by nerve fibres. In midday gerbil the cardiac plexus

consists of about 20 small ganglia, mainly on the right atrium, whereas in chinchilla laniger and Egyptian spiny mouse there are only three to four bigger ganglia and a few smaller ones. In the detailed topography and morphology there were differences among Egyptian spiny mouse, chinchilla laniger and midday gerbil.

In Egyptian spiny mouse and chinchilla laniger, cardiac ganglia were localized mainly on the ventral surface of right atrium, and in midday gerbil on the dorsal surface of left atrium.

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